

Thomas VAUDESCAL

QUANTITATIVE DEVELOPER | FINANCIAL ENGINEERING CONSULTANT

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Professional Profile

Quantitative developer and financial engineering consultant combining **quantitative finance, custom software development and applied AI**. Builds pricing engines, backtesting platforms, automated risk workflows and client-facing business applications using **Python, C++ and Rust**. Experience spans custom software and AI product development at Synerqo, model validation for Canadian financial institutions at Deloitte, and investment strategy development at National Bank Financial Markets. Works with business, risk and trading teams to deliver tools with measurable operational impact. MSc in Financial Engineering, HEC Montréal.

Professional Experience

Synerqo

Montreal, Canada

Quantitative Researcher

May 2025 - Present

- Developed a **custom CRM and advisor portal for an insurance-sector client**, integrating commission processing, electronic signatures and sales dashboards, streamlining business processes and **saving the client tens of thousands of dollars**.
- Designed, built and deployed **Ganesha**, Synerqo's AI knowledge assistant for professional services firms, combining retrieval-augmented generation (RAG) over internal documents, answers linked to their sources, document generation and workflows tailored to each client's methods.
- Built an **event-driven options backtesting engine in Python and Numba**, modeling spreads, slippage, commissions, margin and assignment to validate strategies alongside QuantConnect before live deployment.
- Built an **exotic options and structured products pricing engine** with Monte Carlo, PDE solvers, Greeks via **Adjoint Algorithmic Differentiation (AAD)**, automated hedging, and Heston, SABR and SVI calibration.
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, with latency profiling against the Python research stack.
- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, translating MSc research on dealer gamma exposure into a production signal and execution workflow.
- Technologies:** Python, C++, Rust, TypeScript, React/Next.js, FastAPI, PostgreSQL, QuantLib, QuantConnect.

Deloitte

Montreal, Canada

Quantitative Analyst | Financial Engineering & Modeling - Financial Risk

October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions under IFRS 9 and Basel III/IV frameworks.
- Led independent model validation engagements, assessing assumptions, calibration and discriminatory power through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated monitoring dashboards and validation pipelines in Python**, reducing manual review time by **40%**.
- Authored technical validation reports and regulatory documentation, presenting findings to **senior stakeholders and client risk committees** in the context of OSFI requirements.
- Technologies:** Python, R, SQL, statistical modeling, model monitoring and validation.

National Bank Financial Markets

Montreal, Canada

Quantitative Developer Intern | Global Equity Derivatives R&D

May 2023 - December 2023

- Delivered **\$1M in annual cost savings** through strategy optimization and migration from an external vendor to internal production systems.
- Built an **end-to-end Python backtesting framework** covering signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, implementing constrained portfolio optimization with cvxpy.
- Worked with **structuring and trading teams** to integrate quantitative signals into derivatives pricing and hedging workflows and delivered performance and risk reports to the trading desk and portfolio managers.
- Technologies:** Python, cvxpy, SQL, Bloomberg Terminal & API, AWS (EC2, S3), statistical testing.

Videns Analytics

Montreal, Canada

Data Analyst

April 2020 - June 2021

- Led **consulting engagements in insurance analytics**: data cleaning, exploratory analysis, predictive modeling and automated reporting to support client decision-making.
- Designed and delivered a **Python, statistics and machine learning curriculum** for Videns Academy, adopted by UQAC for a university course. Promoted from intern to full-time Data Analyst.

Technical & Consulting Skills

Development	Python, C++, Rust, SQL, R, MATLAB, VBA, object-oriented programming, asynchronous execution, event-driven systems, latency profiling.
Python	NumPy, Numba, pandas, SciPy, statsmodels, scikit-learn, PyTorch, cvxpy, QuantLib, asyncio.
Quant Finance	Options and structured products pricing, Greeks via AAD, Monte Carlo simulation, PDE methods, volatility calibration (Heston, SABR, SVI), backtesting, portfolio optimization and market microstructure.
Risk Modeling	Credit risk (PD, LGD, EAD), model validation, statistical backtesting, sensitivity analysis, stress testing, model monitoring, IFRS 9, Basel III/IV and OSFI documentation.
Data & Tools	Bloomberg Terminal & API, AWS (EC2, S3), Docker, Git, Linux/Shell, REST APIs, WebSocket, large-scale data processing.
Web	TypeScript, React, Next.js, Node.js, Tailwind CSS.
AI & Apps	RAG, LLM integration, Qdrant, FastAPI, PostgreSQL, SharePoint integration, custom CRM workflows, document generation.
Consulting	Client-facing quantitative analysis, technical documentation, presentation of model findings, collaboration with trading and risk teams, training and knowledge transfer.

Education

HEC Montreal

MSc in Financial Engineering - Thesis

Montreal, Canada

August 2022 - December 2024

- **Thesis:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market. Awarded **A+ with commendation from the examination committee**. Research on dealer hedging and intraday volatility dynamics, subsequently applied to a live systematic options strategy.
- **GPA:** 3.88/4.3. Recipient of the Deloitte M.Sc., Alpha Fixe Capital, Jocelyne & Jean C. Monty I, and Jocelyne & Jean C. Monty II scholarships of excellence.
- **Distinctions:** ConUHacks VIII, National Bank challenge winning team (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon, and 2nd place at the HEC Montreal Trading Club Trading Competition.
- **Leadership:** Vice President Research at the Trading Club and R&D Analyst at the BNI-HEC Investment Fund (\$5M AUM), conducting equity research and portfolio analysis.
- **Selected coursework:** Stochastic Calculus, Derivatives Pricing, Financial Econometrics, Numerical Methods, Machine Learning for Finance, Risk Management.

HEC Montreal

BBA | Specialization in Mathematics, Finance and Economics

Montreal, Canada

August 2017 - May 2022

- **GPA:** 3.8/4.3 for specialization courses. Quantitative track combining mathematics, economics and finance.
- **Leadership & teaching:** Vice President Education at the Data Science Committee, organizing Python workshops and machine learning seminars, and Teaching Assistant for Intermediate Microeconomics (200+ students).

University of Montreal

Minor in Mathematics

Montreal, Canada

August 2018 - August 2022

- Completed alongside the BBA. Probability Theory, Mathematical Statistics, Stochastic Processes, Linear Algebra, Multivariable Calculus, Real Analysis, Programming (C/C++).

Languages

French	Native
English	Professional proficiency (spoken and written)

Interests

Sports	Track and field, marathon running, mountain hiking, indoor climbing, swimming, weight training, scuba diving.
Intellectual	Chess (2000 Elo), mathematics, philosophy, programming.